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Temperature and mass-flow behavior of a He-4 Joule-Thomson cryocooler

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ABSTRACT

The last stage of a He-4 Joule-Thomson cryocooler, which is composed of a recuperator, a JT valve, and a cold heat exchanger, is investigated. It is found that, for given precooling temperature and high and low pressures, the JTC can work steadily with any heat load less than a certain maximum. When the heat load exceeds this maximum the temperature of the last stage rises without limitation. This situation is defined as the overloaded condition. With pressure-enthalpy diagram analysis, the temperature behavior is explained theoretically and verified by the experimental results. The mass flow change along with the upstream state of the JT valve is analysed with the theory of the two-phase choking.

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1. Introduction

Nowadays, precooled Joule-Thomson cryocoolers (PJTCs) are the main cryocoolers at liquid helium temperature for long-life space missions [1,2] due to their advantages, such as high reliability, long lifetime, and low level of vibration and electromagnetic interferences [3,4]. They are mainly applied to cool detectors or to precool other cryocoolers which can achieve even lower temperatures. Under contracts with NASA [5–8], JAXA [9–12], and ESA [13–16], etc., some research institutions and companies have reported their developed PJTCs [17].

Based on thermodynamic analysis, de Waele [18] pointed out that there is an intrinsic cooling-power limitation for Joule-Thomson cryocoolers (JTCs). The temperature will increase unlimited if the heat load, added on the cold heat exchanger (CHX), exceeds a certain critical value which is defined as the maximum cooling power, $\dot{Q}_{max}$. With a heat load $\dot{Q}$ less than $\dot{Q}_{max}$, the JTC can operate steadily. So far a comparison between theory and experiment for these phenomena is missing [18].

For the goal of developing a PJTC prototype and verifying the related theoretical analysis [19], a setup of an open-cycle JTC, precooled by a GM cryocooler, has been built at Zhejiang University [20]. In this paper, the last stage of the JTC, which is composed of a recuperator, a JT valve, and a CHX, is analysed. A pressure-enthalpy (p-h) diagram is applied to explain the temperature behavior when $\dot{Q}$ increases from zero to a value exceeding $\dot{Q}_{max}$. This theoretical analysis is verified by the experiment. The mass flow $\dot{m}$ change along with the upstream state of the JT valve will be explained theoretically under the two-phase choking condition.

2. *p-h* diagram analysis on the last stage

2.1. Last stage

Fig. 1 represents the JTC which is precooled by a two-stage GM cryocooler. The system has been described in a previous paper [20]. Now we focus on the last stage which composes of a recuperator (CFHX3), a JT valve, and a CHX (H3 attached). The He-4, from the high-pressure bottle, is cooled down through CFHX1, PreHX1, CFHX2, and PreHX2 before it flows into the last stage. The gas which flows out from the last stage precools the coming gas in CFHX1 and CFHX2 before it is released to atmosphere. The thermometers are labeled as small squares, numbered from 1 to 17, in Fig. 1. For the last stage some state points are labeled in the order of the flow direction as states a, b, c, d, and e (see the last stage in Fig. 1) whose temperatures are monitored by thermometers 12, 13, 15, 14, and 11 respectively.

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Nomenclature

A_{ch}	characteristic area (m^2)
G	mass flux (kg/m^2-s)
h	specific enthalpy (J/kg)
$\dot{m}$	mass flow (kg/s)
p	pressure (Pa)
q	specific heat load or cooling power (J/kg)
$\dot{Q}$	heat load or cooling power (W)
s	specific entropy (J/kg-K)
T	temperature (K)
V	velocity (m/s)

Greek letters

Δ	difference
ρ	density (kg/m^3)

Superscript

*	choking state

Subscripts

0	stagnation state
a-e	state points
max	maximum

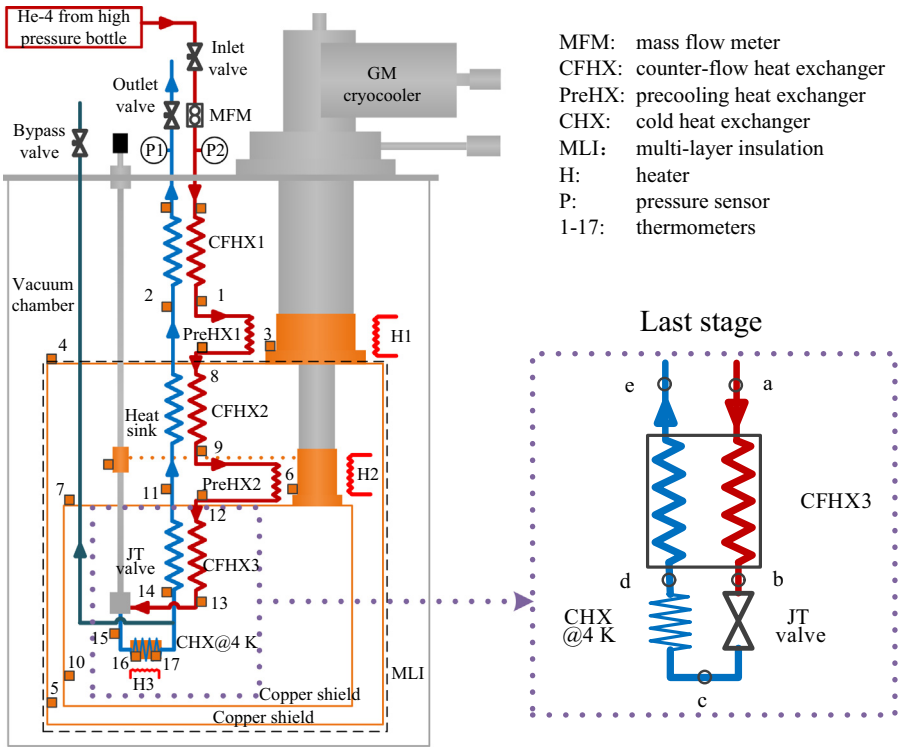


Fig. 1. The open-cycle PJTC experiment setup.

The CFHX3 is a Linde type [21] (tube-in-tube) heat exchanger with the length of 3.3 m. It is composed of an inner tube (1/8 in. standard stainless-steel tube) and an outer tube (1/4 in. standard stainless-steel tube) which both have a wall thickness of 0.035 in. (0.89 mm). The CHX is made of 1/8 in. standard stainless-steel tube which is coiled and soldered around a copper cylinder. The length of the tube, available for heat exchange, is about 0.6 m. The heat load is added to the copper cylinder by an electrical resistance (H3 in Fig. 1) for measuring the cooling power. The JT valve is a needle valve whose flow area can be adjusted. As shown in Fig. 1 the regulating stem of the JT valve has a heat sink cooled by the second stage of the GM cryocooler for reducing the heat leak from room temperature to liquid-helium temperature.

2.2. p-h diagram analysis

The last stage of the JTC will be analysed in this section with a *p-h* diagram similar to our previous work [19]. All the numerical data for He-4 are derived from EES [22] which uses the same equation of state [23] as Refprop 9.1 [24]. In the *p-h* diagram, the thermodynamic processes in the last stage can be described with straight lines. Some assumptions are made to simplify the problem:

1. The CFHX3 is perfect;
2. The mass flow, $\dot{m}$, is constant;
3. Pressure drops only in the JT valve;
4. There is no heat transfer between the last stage and ambient.

The precooling temperature (the temperature of state a, T_a) is fixed at 17.3 K while the high and low pressures are fixed at 1.137 MPa and 0.104 MPa respectively corresponding to the experimental results presented in the next section. Thus state a is fixed on the *p-h* diagram. The heating power, applied to H3, is $\dot{Q}$. We introduce the specific cooling power $q = \dot{Q}/\dot{m}$. When $q = 0$ the thermodynamic processes in the last stage can be plotted as Fig. 2. As

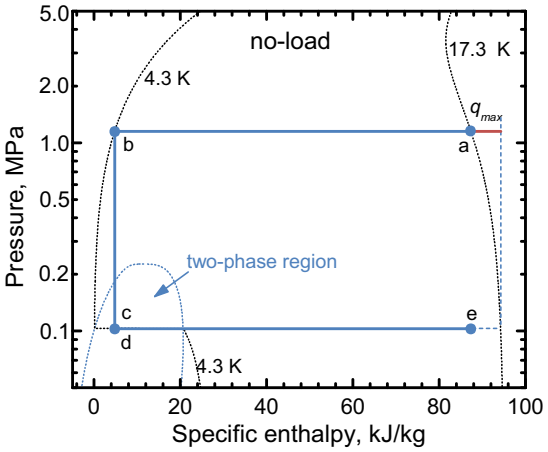


Fig. 2. *p-h* diagram for no-load state. Two isothermal lines are labeled as 4.3 K and 17.3 K. The labels a, ..,e refer to points in the last stage as indicated in Fig. 1.

in Refs. [18,19], the specific maximum cooling power $q_{max,}$ defined as $q_{max} = \dot{Q}_{max}/\dot{m}$, is indicated by the length of the line segment between the state a and the vertical dash line in Figs. 2 and 3. The vertical dash line is the isenthalp whose enthalpy value is determined by the low pressure and T_a.

The temperature of state b, T_b, turns out to be equal to 4.3 K. When the He-4 throttles through the JT valve from state b to state c, it enters the two-phase region along the isenthalpic line. As $T_b \approx T_c$ the throttling process is not only isenthalpic but also isothermal. State d is the same as state c because $q = 0$. After that the low-pressure He-4 precools the high-pressure flow in the CFHX3 and arrives at state e. The enthalpy difference between states a and b, Δh_{ab}, should be equal to that between e and d, Δh_{ed}, according to the first law of thermodynamics. Between states e and a, $\Delta h_{ea} = 0$ since $q = 0$.

When the heat load is increased gradually, states d and e will move to the right along the isobar in the *p-h* diagram (see Fig. 3) while the states a, b, and c keep their original positions. The temperature T_e rises while T_d does not change because state d is still in the two-phase region. If the specific heat load $q = q_{max}$, T_e will be the same as the precooling temperature, T_a. Then, the enthalpy difference $\Delta h_{ea} = q_{max}$.

If the heat load keeps on increasing from the steady condition of Fig. 3, state d will keep on moving to the right but state e will not

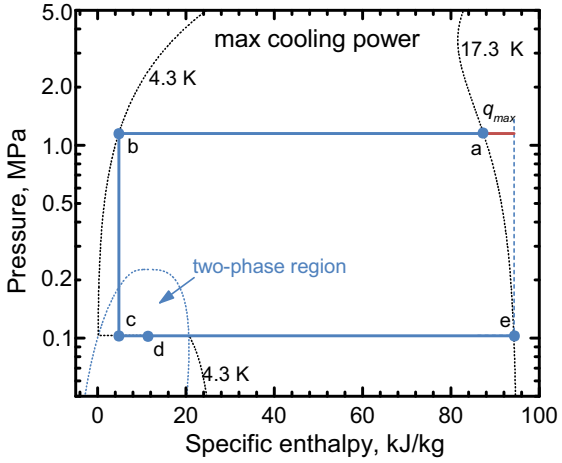


Fig. 3. *p-h* diagram for maximum cooling power state.

because T_e cannot exceed T_a. It makes Δh_{ed} decrease and the high pressure gas in the CFHX3 cannot be precooled sufficiently. Thus states b and c will also move to the right along the isobar on the *p-h* diagram. As long as $q > q_{max}$, the JTC will never come back to a steady condition. As state b is single phase, T_b is the first to rise away from liquid-helium temperature, while T_d is the second and T_c the last. This non-steady condition, happening when $q > q_{max}$, is defined as overloaded condition.

The *p-h* diagram analysis shows that the JTC can operate steadily if $q \le q_{max}$. Especially when the CHX contains two-phase helium, its temperature will not change as the heat load increases. This is a nice characteristic of a JTC and distinguished from regenerative cryocoolers [25–29] of which the temperature always rises with the heat load. The analysis also explains why the JTC cannot work steadily when the $q > q_{max}$. With an efficient recuperator, the throttle process can be both isenthalpic and isothermal. In the next section, experimental results will be presented to verify this *p-h* diagram analysis.

3. Experimental results

The setup is shown in Fig. 1. The details of the sensors are listed in Table 1. These Cernox thermometers were calibrated by the Technical Institute of Physics and Chemistry, Chinese Academy of Sciences, with the accuracy of 0.1 K. The thermometers were mounted inside small copper blocks which were soldered on the connecting tubes. Apiezon N grease was used to provide good thermal contact. Helium-4 with high purity of 99.9999% was used.

In the experiment the inlet temperature T_a was fixed at around 17.3 K by adjusting H1 and H2. The high pressure was kept at 1.137 MPa while the low pressure was 0.104 MPa which is close to one atmosphere. The voltage and current, applied to H3, were measured with the four-wire method, two leads for voltage and another two for current. The temperatures of states a to e are plotted in Fig. 4. The time line is divided into regions A, B, C, D, E, F, G, and H by vertical dashed lines. The heat load, applied for each region, is also given in Fig. 4 in mW.

Under the steady condition when $\dot{Q} = 0$ (see region A), the temperature T_b was close to T_c and T_d at around 4.4 K indicating that the CFHX3 had a good efficiency. There is a small difference between the experimental data (4.4 K) and the theoretical value (4.3 K in Section 2.2) which can be caused by the pressure drop of the low pressure flow in the CHX and the three CFHXs. It is ignorable considering the small difference is within the measuring accuracy (±0.1 K). Meanwhile, $T_e < T_a$. All the temperatures behaved the same as Fig. 2. When $\dot{Q}$ was increased to 39.8 mW (region B) and 48.9 mW (region C), the JTC worked steadily with T_b, T_c, and T_d constant while T_e was getting closer to T_a in agreement with Fig. 3.

In the regions D, F, and H heat loads of 58.8, 52.9 and 49.4 mW were applied. All of them made the JTC overloaded. The temperature T_b was the first to rise away from 4.4 K, and T_d the second which also agrees with the *p-h* diagram analysis. These results

Table 1
Details of the measuring sensors.

Sensors	Corresponding abbreviation in Fig. 1	Brand/type
Pressure sensors	P1 and P2	GE UNIK 5000
Mass-flow meter	MFM	Bronkhorst F-111B
Thermometers	11,12,13,14 and 15	Cernox 1050

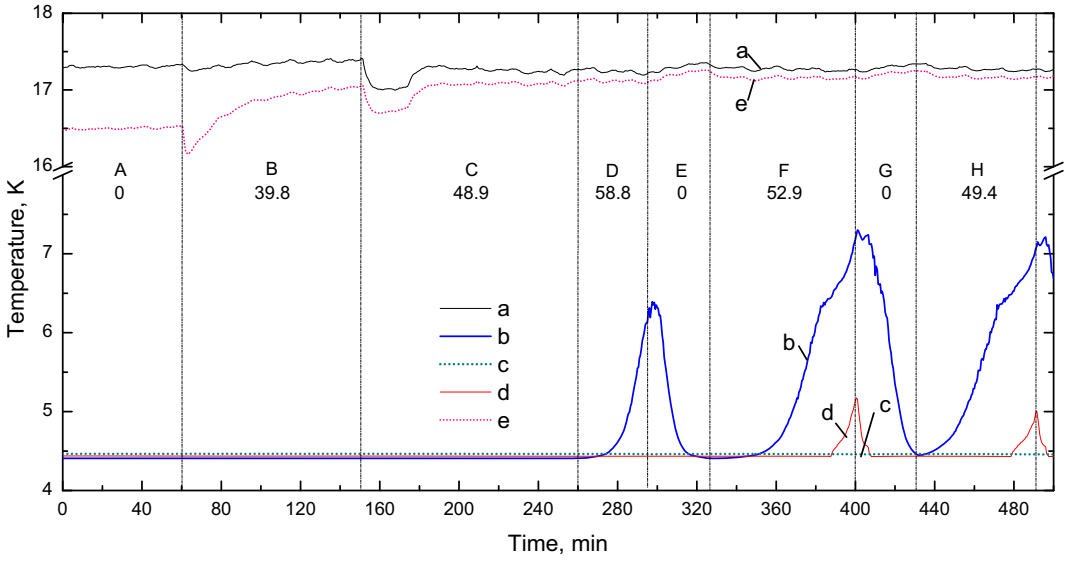


Fig. 4. temperature behavior of the last stage as a function of time. Each region is labeled with the applied heat load as mW.

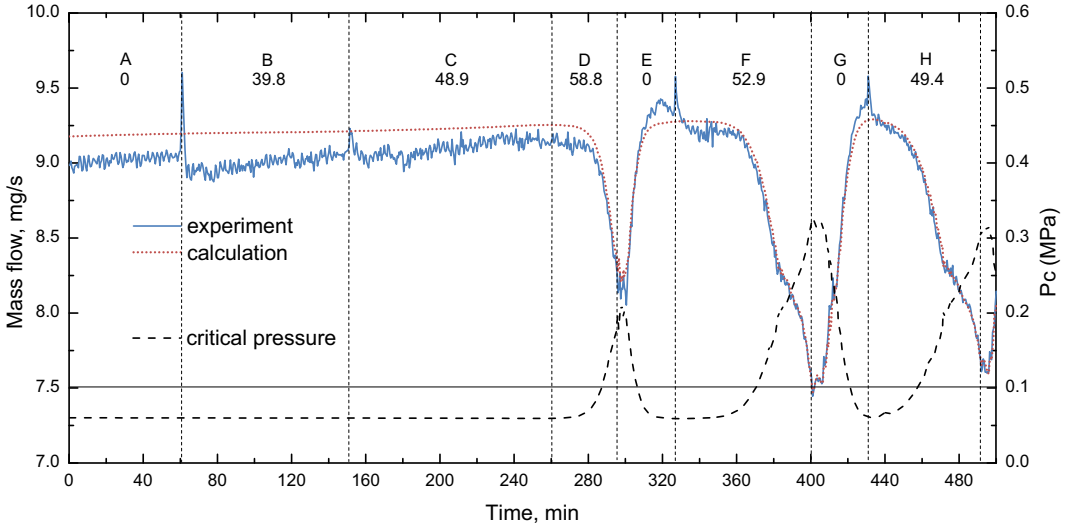


Fig. 5. Mass flow behavior and critical pressure for choking condition.

show that $\dot{Q}_{max}$ is between 48.9 and 49.4 mW. With the experimental data in region C, the calculated cooling power [22] is

$$\begin{aligned}\dot{Q} &= \dot{m}(h_e - h_a) \\ &= \dot{m}[h(0.104\ \text{MPa}, 17.1\ \text{K}) - h(1.137\ \text{MPa}, 17.3\ \text{K})] \\ &= 9.1 \times (93.17 - 87.40)\text{mW} = 52.5\ \text{mW}\end{aligned} \quad (1)$$

The difference between the calculated and the measured cooling power is only 3.6 mW which may be due to nonzero heat leaks, inaccuracy of the measurement, and deviations of the equation of state.

The overloaded condition is accompanied by a mass-flow drop shown as the blue[1] solid line in Fig. 5. The changes of T_b and the mass flow $\dot{m}$ started and stopped simultaneously. The mass flow decreased as soon as the T_b increased. Now we will propose an explanation for this mass-flow behavior.

[1] For interpretation of color in Fig. 5, the reader is referred to the web version of this article.

4. The mass-flow behavior

The decrease of the mass flow accelerates the process of the JTC becoming unstable because it degrades the cooling power. Although the intrinsic limitation of JT cryocooler, as discussed in Sections 2 and 3, is the main reason for the overload, the mass-flow drop also needs attention.

Usually a JTC, working at liquid helium temperature, has a pressure ratio larger than the critical pressure ratio for choked flow [21]. Maytal et al. [30,31] have given the method to predict the critical mass flow for a real gas whether it is in the single- or in the two-phase condition. The JT valve, used in our experiment, is an adjustable needle valve. According to Section 6.1 of Ref. [21], it is reasonable to consider the adjustable JT valve as a circular short duct which has no upstream dissipation. Although the structure of an adjustable needle valve is more complex and the upstream dissipation may exit in a real case due to the complex flow regime near the throttle, the effects of them are still unclear for the JT valve in our case. Under this situation, the assumption

that the expansion process in the upstream preserves entropy is made here to simplify the problem. So that,

$$s(p,T) = s_0 = s(p_0,T_0) = \text{Constant} \tag{2}$$

where s is for specific entropy while p represents the pressure and T the temperature. The subscript 0 stands for the stagnation state. Considering the small deviation of cooling power mentioned in Section 3, the heat leak between the fluid and the surroundings is supposed to be negligible. Thus energy conservation requires

$$h(p,T) + \frac{1}{2}V^2 = h_0 = h(p_0,T_0) = \text{Constant} \tag{3}$$

where V is the velocity. Combining Eqs. (2) and (3), the mass flux, which is assumed uniform on the cross section, can be calculated as function of the pressure

$$G(p) = V(p)\rho(p) \tag{4}$$

where ρ stands for density. As introduced by the Maytal et al. [31], the critical mass flux, G^*, is determined by maximizing G with respect to pressure. The pressure which maximizes G determines the choking state, (p^*, T^*), for any given stagnation conditions, (p_0, T_0).

The velocity at the inlet of the JT valve (state b) is practically zero (0.025 m/s). So state b, will be used as the stagnation condition, (p_0, T_0). The regions F and G in the Fig. 5 are combined to an example region for the following analysis.

As shown in Fig. 6, with the experimental results of state b, the calculation results of critical mass flux G^* has the same tendency with $\dot{m}$ in Fig. 5. The characteristic area A_{ch} presented in Fig. 6 is defined as the ratio of the measured mass flow and the calculated G^*, namely

$$A_{ch} = \dot{m}/G^* \tag{5}$$

which should be regarded as a parameter that includes the actual effects of the friction, heat transfer, shape of the JT valve, etc.

In Fig. 7 A_{ch} is plotted as the function of G^* which shows a linear relationship. The fitting correlation is

$$A_{ch} = (85436 - 1.7668G^*) \times 10^{-14} \tag{6}$$

with A_{ch} in m^2 and G^* in kg/m^2-s.

It is shown that A_{ch} decreases linearly as G^* increases. The reason for the variation of A_{ch} with G^* is not clear at the moment. A CFD model is thought to be helpful to find out the main factors but this work will be considered in the future. Eq. (6) can be used as an empirical correlation. As G^* can be calculated using the experimental data of state b for all the regions on the Fig. 5, the mass flow $\dot{m}$ is reproduced as the dotted line combining Eqs. (5) and (6). The calculation result matches the experiment rather well especially when the critical pressure p^* is higher than the low pressure (see the isobar 0.104 MPa in Fig. 5). Then the applicability of Eq. (6) has been further verified and it shows a method to get the characteristic area of a JT valve. The regions, in which p^* is below 0.104 MPa, theoretically do not match the choking condition and the mass flow should be determined by the upstream and downstream states of the JT valve. Based on the above analysis, the mass-flow decrease accompanied by the temperature rise can be explained by the theory of two-phase choking.

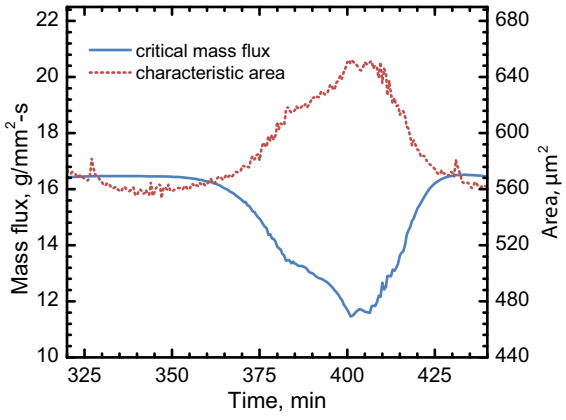


Fig. 6. Calculation of critical mass flux and characteristic area.

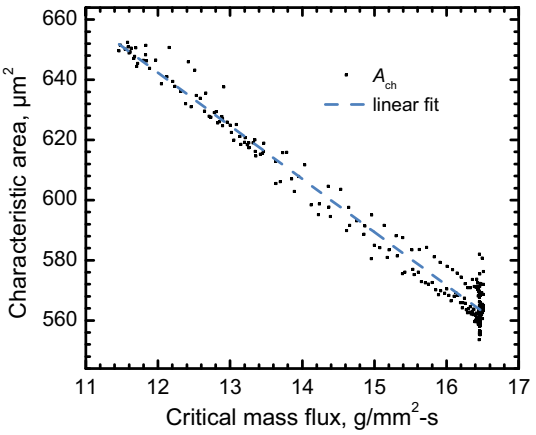


Fig. 7. Fitting results between critical mass flux and characteristic area.

5. Summary

The thermal behavior of the JTC last stage has been illustrated with the *p-h* diagram analysis and verified with the experimental results. It shows that the Joule-Thomson cryocooler can provide cooling power at a constant temperature up to a certain maximum value. With a well-designed recuperator (CFHX3), the JT expansion is both isenthalpic and isothermal ($T_b = T_c$).

The mass-flow behavior is discussed with the choked-flow theory for two-phase condition. It is the first time to apply this theory to the condition in which the stagnation state of He-4 is in the subcooled liquid state. Although the reason for the varying characteristic area is not clear, the characteristic area fitting correlation with respect to the critical mass flux explains the experimental results very well.

In the future the cooling power will be studied more systematically as function of the high pressure and the precooling temperature.

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